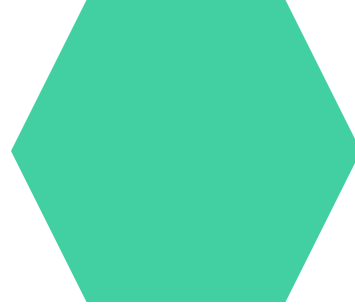
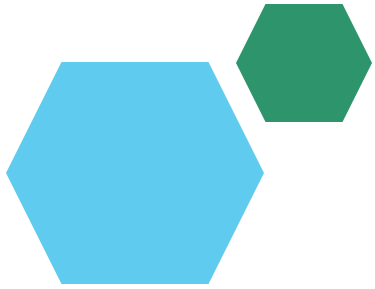




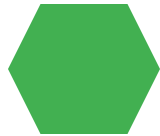
# Project Title - Seasonal Agricultural Performance Analysis

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**COLLEGE NAME-** School Of Management Sciences, LUCKNOW

**AICTE STU ID :**

STU690cb34e4f5eb1762440014



# PROBLEM STATEMENT

- (Agricultural activities are influenced by seasonal variations in environmental conditions, farming practices, resource availability and market conditions. As a result, agricultural performance may differ from one season to another.
- However, raw agricultural data does not clearly explain how agricultural performance changes across seasons or what patterns can be observed in different seasonal conditions.
- The problem is to analyze the given agricultural dataset and investigate seasonal differences in agricultural performance by identifying meaningful patterns, trends, relationships and variations within the available data.



## Project Description - Seasonal Agriculture Performance Analysis

Seasonal Agriculture Performance Data Analytics is a data analysis project that studies agricultural performance across different crops, seasons, regions, irrigation methods, environmental conditions, and resource usage. The project uses data analytics and visualization techniques to identify patterns and relationships that influence crop productivity.

The analysis focuses on comparing crop yields across states and seasons, understanding irrigation and water efficiency, examining resource utilization, and identifying variations in agricultural performance.

Different charts and statistical analyses are used to convert the agricultural dataset into meaningful insights.

The main objective of this project is to understand seasonal and regional agricultural trends and identify factors associated with better crop performance, which can support data-driven agricultural planning and efficient resource management.

Key areas covered:

- 🌱 Crop-wise performance analysis
- 🌦️ Seasonal yield patterns
- 📍 Regional/state-wise differences
- 💧 Irrigation and water efficiency
- 🌱 Resource utilization
- 📊 Data visualization and exploratory analysis
- ⚠️ Identification of agricultural performance patterns and risks

# WHO ARE THE END USERS?



The main end users of the **Seasonal Agriculture Performance Data Analytics** project are:

- **Farmers** - To understand which crops and seasons show better performance and make informed farming decisions.
- **Agricultural Officers** - To monitor agricultural productivity and identify regional or seasonal variations.
- **Government & Policy Makers** - To support data-driven agricultural policies, planning, and resource allocation.
- **Agricultural Researchers** - To study crop, seasonal, environmental, and resource-use patterns.
- **Agribusiness Companies** - To understand crop production trends and plan procurement and supply chains.
- **Irrigation & Water Management Authorities** - To analyze irrigation methods and improve water-use efficiency.
- **Students & Data Analysts** - To explore agricultural datasets and develop data-driven insights.

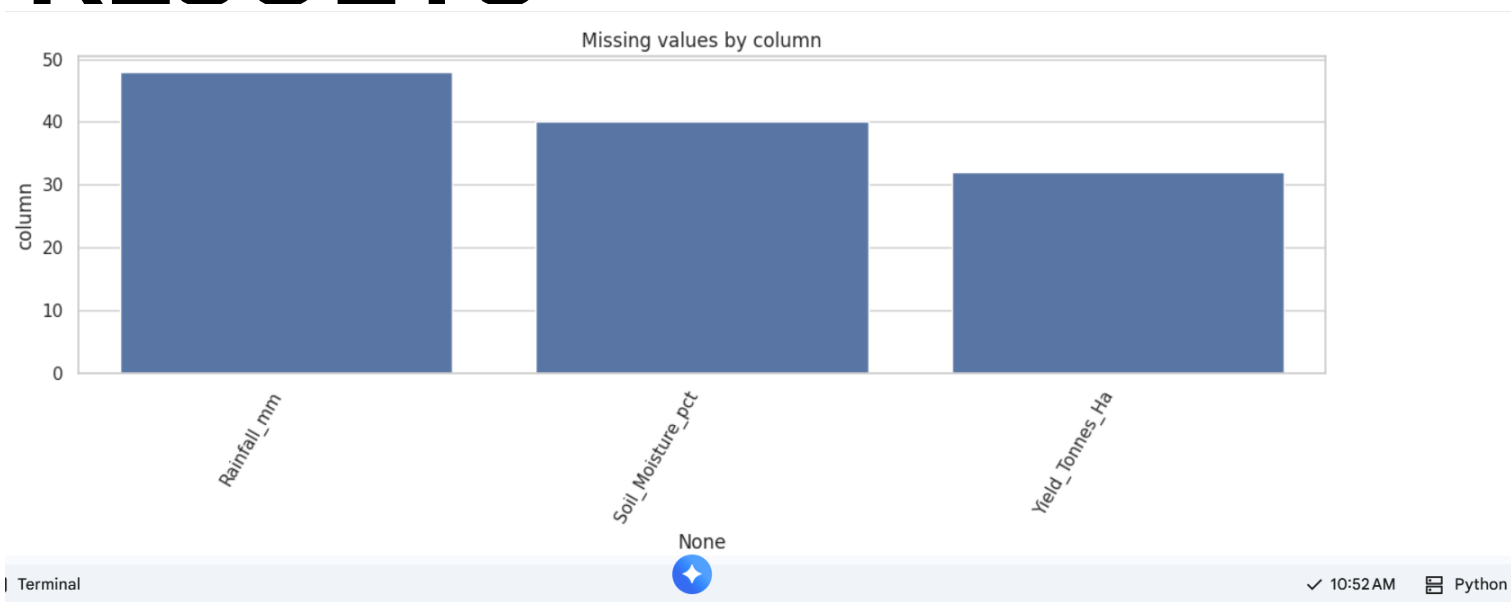


# Technology Used

Python, Pandas, Matplotlib, Seaborn and Exploratory Data Analysis (EDA).



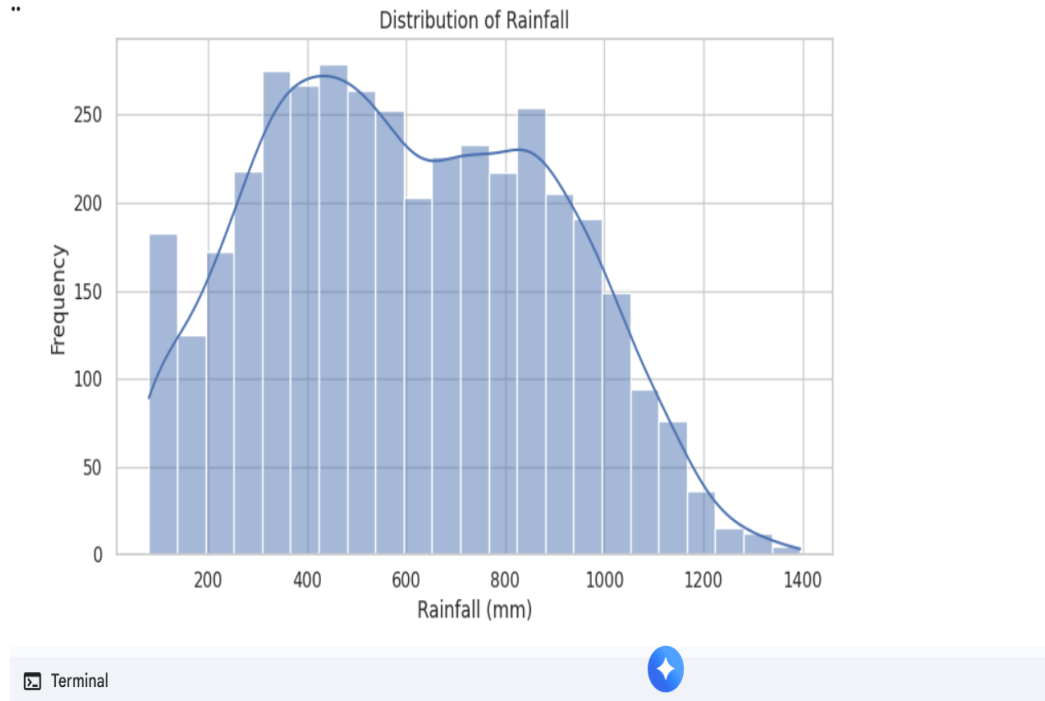
# RESULTS



## Three columns have missing values:

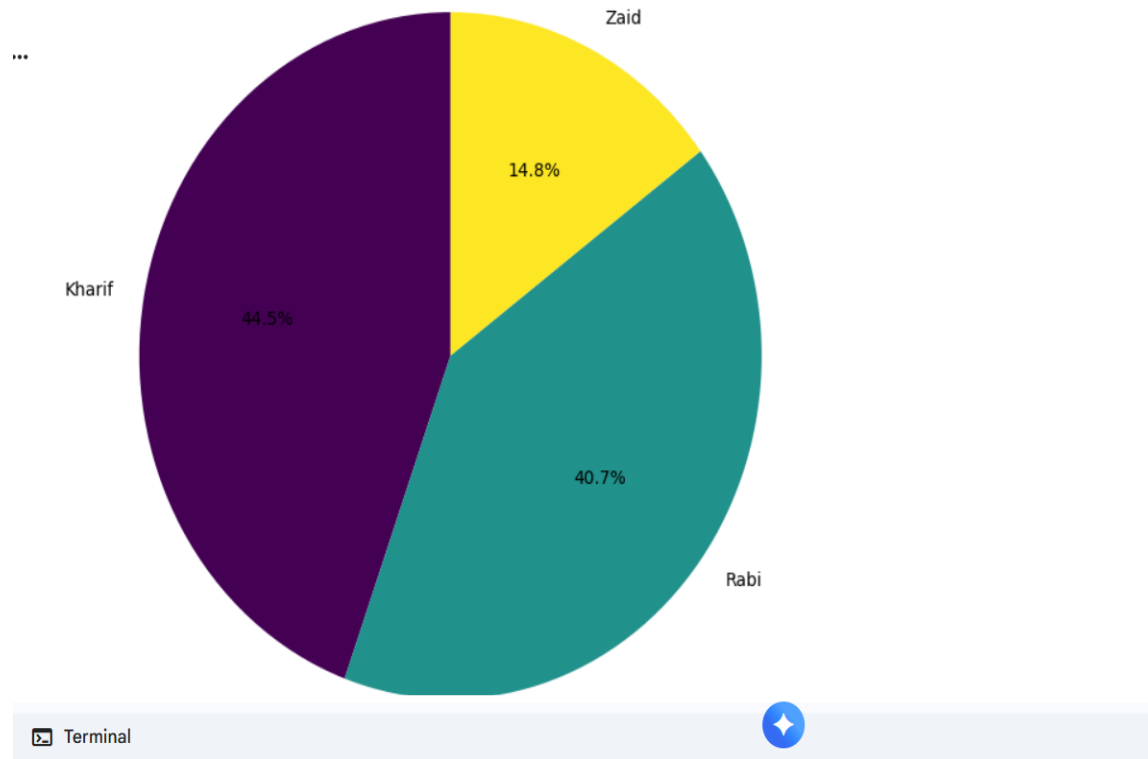
The graph displays three distinct bars, meaning there are exactly three columns in your dataset that contain missing data (where the count is greater than zero). Distribution of missing data: The first column has the highest amount of missing data, with approximately 48 missing values. The second column has around 40 missing values. The third column has the lowest among them, with about 32 missing values.

# RESULTS



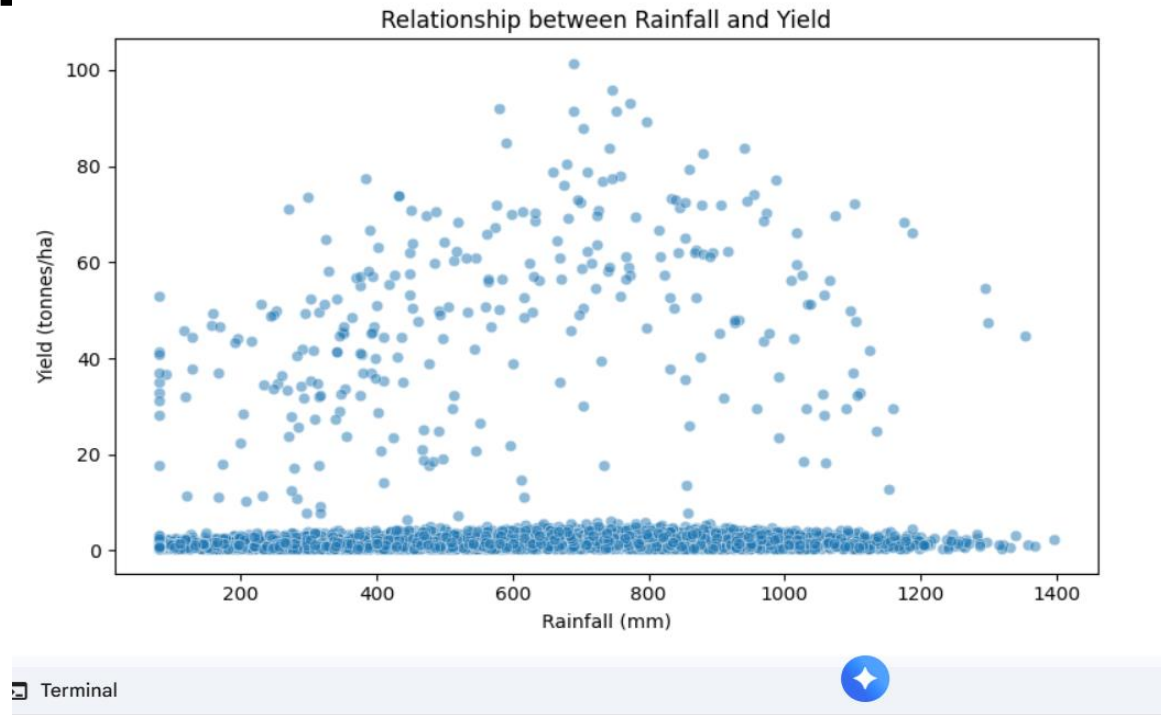
The graph shows two prominent peaks (modes) in the distribution. The highest peak (primary mode) occurs around higher frequency counts near the top gridlines. A secondary peak (smaller mode) is visible further down the scale before tapering off.

# RESULTS



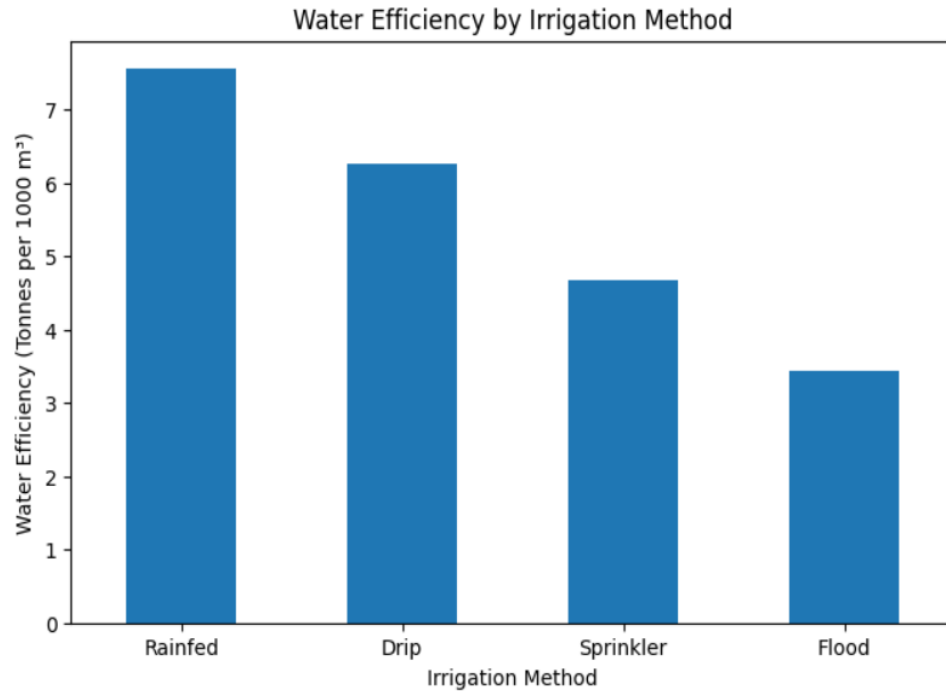
The data is highly unevenly distributed among the seasons. The Kharif season has the highest number of recorded entries, followed by Rabi. The Zaid season has the fewest records, making up less than 15% of the entire dataset.

# RESULTS



This pattern suggests that excessive rainfall may negatively impact this specific crop. This could be due to factors like waterlogging, soil erosion, nutrient leaching, or increased disease prevalence in overly wet conditions.

# RESULTS



The data highlights that targeted or natural watering methods (Rainfed and Drip) are vastly superior in water productivity compared to broad-distribution methods (Sprinkler and Flood), which lose significant amounts of water to evaporation and runoff.

# Future scope

The **Seasonal Agriculture Performance Data Analytics** project can be further developed into a more advanced agricultural decision-support system. Future improvements can include:

- 1.Crop Yield Prediction** - Use Machine Learning models to predict crop yield based on historical and seasonal data.
- 2.Weather Integration** - Integrate real-time weather data such as rainfall, temperature, and humidity to study their impact on crop performance.
- 3.AI-Based Crop Recommendation** - Develop a system that recommends suitable crops based on soil, climate, season, and available resources.
- 4.Real-Time Monitoring** - Connect IoT sensors to monitor soil moisture, temperature, and other field conditions.
- 5.Water Management** - Build an AI-based irrigation recommendation system to improve water-use efficiency.
- 6.Interactive Dashboard** - Create a Power BI, Tableau, or web-based dashboard for farmers, researchers, and government authorities.
- 7.Early Risk Detection** - Predict possible low-yield periods or agricultural risks using historical and environmental patterns.
- 8.Mobile Application** - Develop a mobile app that provides farmers with crop insights, recommendations, and alerts in an easy-to-understand format.

# GitHub Link

<https://github.com/vaibhavishukla45-code/Seasonal-Agricultural-Performance-Analysis-.git>



# VOIS Course completion certificate - Course name- Data Visualization



# Thank you